The Digital Divide in AI Governance: How ICT Sector Development Influences National AI Strategy Adoption

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Abstract

As national AI strategies have become a common feature in national policy agendas, notable disparities have emerged between adopters and non-adopters. This paper asks whether a country's ICT sector development—measured principally via ICT service exports—independently predicts AI strategy adoption, the timing of adoption, and the volume of AI-related legislation, or whether the apparent relationship is better explained by broader structural conditions. Drawing from a cross-national dataset of 128 countries assembled from the Stanford AI Index, IMF AI Preparedness Index, the ITU Internet Development Index, and the World Bank's World Development Indicators, the analysis employs logistic, OLS, and Poisson regression to model each outcome.

ICT service exports are positively associated with strategy adoption in univariate specifications, but this association does not survive the introduction of controls for economic development, population, and digital infrastructure. Across all three specifications, GDP per capita emerges as the most consistent predictor, with population and broadband penetration also playing significant roles. These patterns hold when predictors are re-estimated using 2016 baseline values measured prior to widespread strategy adoption, providing a partial check against reverse causality. The findings suggest that the global AI governance gap is real, but does not primarily reflect ICT sector strength—it reflects the same foundational conditions of economic development and state capacity that underpin longstanding inequalities between the Global North and South in tech, with corresponding implications for how interventions to close that gap might be designed.

Introduction

As artificial intelligence moves from a subject of speculative interest to a central preoccupation of governments worldwide, the question of which countries lead in AI policy adoption—and more importantly, why they do so—has taken on increasing urgency. By the beginning of 2021, thirty-four countries had published national AI strategies; that figure has since more than doubled (Fatima, Desouza, & Dawson, 2020; AI Index Steering Committee, 2026). These strategies, however, have not emerged uniformly. Wealthier, more technologically advanced countries have tended to move first, while much of the developing world remains without formal policy frameworks—a pattern which closely mirrors longstanding digital divides between the Global North and South (UNCTAD, 2006; Cazzaniga et al., 2024). If national AI governance becomes a prerequisite for shaping international standards, accessing AI-driven economic opportunities, and mitigating potential risks, then understanding what drives country adoption—and what leaves others behind—has clear implications for the trajectory of global digital inequality.

A growing body of research has begun to map this terrain. Fatima, Desouza and Dawson (2020) conducted a content analysis of early national AI strategies, documenting how countries conceptualised AI's role in the public sector, approached data governance, and planned capacity-building investments. Their analysis, however, was descriptive rather than explanatory, and did not attempt to identify the conditions under which countries adopt strategies in the first place. Subsequent work by the same authors examined the extent to which national AI plans reflect underlying implementation capacity, finding that the resources required—particularly technological infrastructure and human capital—vary significantly across adopters (Fatima et al., 2022). Elsewhere, Cazzaniga et al. (2024) developed the IMF's AI Preparedness Index across

174 economies, demonstrating that advanced economies score systematically higher on measures of digital infrastructure, innovation, and regulatory readiness. Yet the relationship between these indicators and the specific decision to adopt a national AI strategy remains untested. Ali, Mustafa and Aysan (2025), working with cross-sectional data on generative AI adoption, similarly find that IT infrastructure, R&D investment, and firm-level technological activity predict adoption, while misaligned government policies can hinder it—pointing to a gap between technological capacity and policy outcomes.

What remains underexplored in this literature is whether a country's private-sector ICT capacity—the size and orientation of its information and communication technology industries—independently predicts AI policy adoption, or whether the observed relationship between ICT capacity and governance is better explained by broader economic and institutional conditions. This paper addresses that question directly, asking whether ICT sector development predicts national AI strategy adoption once differences in economic development, population size, and digital infrastructure are taken into account. If ICT sector strength operates as an independent driver, then countries with larger ICT sectors should be more likely to adopt AI strategies, adopt them earlier, and engage more extensively in AI-related legislation, even after controlling for these factors.

To examine this, the paper assembles a cross-national dataset of 128 countries, drawing on the Stanford AI Index (Maslej et al. 2025; Sajadieh et al. 2026), the IMF AI Preparedness Index, the ITU Internet Development Index, and the World Bank's World Development Indicators. It employs logistic regression to model the likelihood of strategy adoption, ordinary least squares regression to model the timing of adoption among adopters, and Poisson regression to model the number of AI-related bills passed into law. Across these specifications, the analysis

tests whether ICT sector development—measured primarily through ICT service exports—predicts AI governance activity once key confounding variables, including GDP per capita, population size, and digital infrastructure, are accounted for.

The findings suggest that the "digital divide in AI governance" is real, but operates differently than a sector-specific account would predict. ICT service exports are positively associated with strategy adoption in univariate specifications, but this association does not persist once controls for economic development and infrastructure are introduced, indicating that the observed relationship is not independent. Instead, GDP per capita emerges as the most consistent predictor across all three specifications, with population and broadband also playing significant roles in the OLS and Poisson models. These patterns hold when re-estimated using 2016 baseline indicators measured prior to widespread strategy adoption, providing a partial check against reverse causality. Taken together, the results suggest that cross-national differences in AI governance are better understood as a function of broader structural conditions than of ICT sector strength alone.

Data

This analysis utilized a cross-national dataset assembled from five principal sources, including indicators of national AI policy activity, measures of ICT sector development, digital infrastructure, human capital, and general economic capacity. Each observation corresponded to a single country, observed cross-sectionally, with indicator values drawn from 2025 or, when 2025 data was unavailable, the nearest available year within a five-year window. After merging and cleaning, the dataset consisted of 128 countries and 19 variables, although total counts differed slightly across each variable due to differences in degree of assessment.

All data assessing AI strategy adoption was drawn from the Stanford AI Index's 2026 and 2025 reports on national AI governance activity, which record whether a government has released a national AI strategy, has one in development, or has adopted no strategy at all, alongside the year of strategy publication and the cumulative number of AI-related bills passed into law for over 100 countries (the most recent data for bills passed is from the 2025 report). The Index is one of the foremost resources on AI strategy adoption and AI-related lawmaking and the most comprehensive publicly available source for this analysis.

For the purposes of this analysis, strategy adoption was operationalised as a binary variable coded 1 where a strategy has been formally released and 0 otherwise; countries with strategies recorded as "in development" were coded as non-adopters, on the grounds that an unreleased strategy did not yet constitute a binding policy commitment. Of the 128 countries retained after filtering, 81 have released a national AI strategy, six have one in development, and 41 have adopted none. Strategy adoption years range from 2017 to 2024, and the number of AI-related bills passed into law is recorded for 113 countries, with a distribution that is heavily right-skewed.

The principal independent variables assessed ICT sector development through three World Bank indicators: ICT service exports as a percentage of total service exports, ICT goods exports as a percentage of total goods exports, and high-technology exports as a percentage of manufactured exports. These variables were intended to measure the extent to which a country's economy is oriented around the production and export of information and communications technologies—a dimension of national capacity that, if the hypothesised relationship holds, should be associated with earlier and more extensive AI policy adoption.

Control variables were drawn primarily from the World Bank's World Development Indicators and were selected to account for plausible confounds that could potentially predict both ICT sector size and AI strategy adoption. These included GDP per capita (PPP-adjusted) and total population controlled for general economic scale and wealth; fixed broadband subscriptions per 100 people and the percentage of individuals using the internet captured digital infrastructure independently of ICT exports; gross domestic expenditure on research and development as a percentage of GDP, resident patent applications, and gross tertiary enrolment ratios served as proxies for national innovation ecosystems and human capital; and government expenditure on education was used to measure public investment in knowledge production. Two composite indices were also incorporated as alternative specifications: the IMF's AI Preparedness Index, a single score measuring regulatory, institutional, and infrastructural readiness for AI across 165 countries, and the International Telecommunication Union's Internet Development Index for 2023, which provided a composite measure of connectivity and digital access across 169 countries.

Coverage varies across indicators (e.g., R&D expenditure available for 91 of 128 countries; ICT goods exports for 113; patent data most recent in 2021). Each World Bank indicator was therefore drawn from the year nearest 2025 within a five-year window, with the nearest available value between 2020 and 2025 used where 2025 data was missing. Aggregate and regional codes were excluded, as were fourteen small territories whose inclusion was not uniform across indicators. To guard against the possibility that current indicator values may have been influenced by strategy adoption itself, a parallel baseline dataset was constructed using 2016 indicator values, predating any country’s publication of a national AI strategy. This baseline served as a robustness check against reverse causality.

Methods

This paper examines whether countries with larger and more developed ICT sectors are more likely to adopt national AI strategies, adopt them earlier, and engage more extensively in AI-related legislation. It hypothesises that ICT sector development—measured through a country's ICT exports—will be positively associated with each of these outcomes. If private-sector technological capacity translates into governmental policy action, then countries with greater ICT service, goods, and high-technology exports should be more likely to adopt a national AI strategy, should do so earlier, and should also exhibit greater legislative activity in this domain. The analysis therefore focuses on whether this relationship holds once broader structural factors are considered.

To test these hypotheses, the analysis employed three regression frameworks mapped to different operationalisations of the dependent variable. The first was a logistic regression modelling the binary outcome of whether a country had released a national AI strategy: $P(\text{strategy adopted} = 1 \mid X) = F(\beta_0 + \beta_1\text{ICT_services} + \beta_2\text{log_GDP} + \beta_3\text{log_pop} + \beta_4\text{broadband})$ where F is the logistic function $F(x) = 1 / (1 + e^{-x})$. The second was an ordinary least squares (OLS) regression modelling the year of strategy adoption among the subset of countries that had adopted one: $\text{strategy_year} = \beta_0 + \beta_1\text{ICT_services} + \beta_2\text{log_GDP} + \beta_3\text{log_pop} + \beta_4\text{broadband} + \varepsilon$, And the third was a Poisson regression modelling the count of AI-related bills passed into law: $\log(\lambda) = \beta_0 + \beta_1\text{ICT_services} + \beta_2\text{log_GDP} + \beta_3\text{log_pop} + \beta_4\text{broadband}$, where $\lambda = E(\text{bills passed} \mid X)$. Together, these three approaches addressed distinct dimensions of the research question: whether ICT sector development predicted the likelihood of adoption, the timing of adoption, and the breadth of legislative activity.

All three analyses follow a common estimation strategy. Each begins with a univariate regression including ICT service exports as the sole predictor. Of the three ICT export indicators available, ICT service exports as a percentage of total service exports is used as the primary variable, as it most directly captures software and digital services capacity, which is plausibly more closely linked to AI policy awareness and strategy formation than hardware-oriented indicators. ICT goods exports, by contrast, are heavily shaped by global supply chain positioning and are concentrated in a small number of countries, making them a less suitable primary measure of broadly distributed technological capacity.

A set of core controls is then introduced, consisting of log GDP per capita, log population, and fixed broadband subscriptions per 100 people. These variables are included to account for key confounding factors that may jointly influence both ICT sector development and AI strategy adoption. Economic wealth, country size, and digital infrastructure are all plausible drivers of both technological capacity and policy adoption; omitting them risks attributing to ICT exports what is in fact the effect of underlying development.

Subsequent specifications assess the robustness of the relationship across alternative control configurations. One specification adds the remaining ICT indicators (ICT goods exports, high-technology exports, and log patent applications) alongside the core controls, while another incorporates measures of human capital and innovation capacity, including tertiary enrolment and R&D expenditure as a percentage of GDP. A final specification includes all variables simultaneously. GDP per capita, population, and patent applications are log-transformed prior to estimation to address substantial right-skew in their distributions.

As a robustness check, all logistic, OLS, and Poisson models are re-estimated using a parallel dataset constructed from 2016 indicator values, predating the publication of any national

AI strategy. This baseline specification provides a partial test against reverse causality by ensuring that predictor values are measured prior to the period in which most countries adopted their strategies. Additional specifications replacing broadband penetration with the IMF AI Preparedness Index and the ITU Internet Development Index are also estimated to assess the sensitivity of the results to alternative measures of digital and institutional readiness. While these approaches strengthen the credibility of the findings, the cross-sectional design limits the extent to which causal claims can be made, and unobserved factors—such as political institutions or regional policy diffusion—may still influence the results.

Results

Descriptive Statistics

Countries that have released a national AI strategy differ from non-adopters on nearly every dimension measured. As summarised in Table 1, mean GDP per capita among adopters is $40,171 compared with $21,594 among countries with no strategy; broadband penetration averages 26.1 subscriptions per 100 people versus 16.9; and R&D expenditure as a share of GDP is nearly twice as high (1.47 percent versus 0.84 percent). The gap in ICT service exports—the principal independent variable—is more modest: 11.8 percent among adopters versus 6.7 percent among non-adopters, a difference that is statistically significant but smaller in magnitude than the gaps in GDP, broadband, or tertiary enrolment. Welch two-sample t-tests confirm significant differences between adopters and non-adopters for all variables except mobile cellular subscriptions ($p = 0.75$) and government education expenditure ($p = 0.07$), suggesting that strategy adoption is broadly associated with higher levels of economic development and digital infrastructure rather than with any single ICT-specific indicator.

Table 1: Descriptive Statistics by AI Strategy Status

	Strategy Status		
	None (n = 41)	In Development (n = 6)	Released (n = 81)
GDP per capita (PPP)	21,594	15,947	40,171
Broadband per 100	16.9	10.8	26.1
Internet users (%)	69.4	71.4	83.3
ICT service exports (%)	6.7	12.8	11.8
High-tech exports (%)	8.8	14.6	16.9
R&D expenditure (% GDP)	0.84	0.34	1.47
AI Preparedness	0.416	0.444	0.577

Note: Values are group means. N = 128 countries.

Logistic Regression

In the univariate logistic analysis, ICT service exports are positively and significantly associated with AI strategy adoption ($\beta = 0.042$, $p < 0.1$), corresponding to an odds ratio of 1.043—each additional percentage point of ICT service exports is associated with a 4.3 percent increase in the odds of having adopted a strategy.

This association, however, does not survive the introduction of controls. Once log GDP per capita, log population, and broadband penetration are added in the core controls specification, the coefficient on ICT service exports falls to −0.002 and becomes statistically indistinguishable from zero ($p > 0.9$). Log GDP per capita ($\beta = 0.725$, $p < 0.1$) and log population ($\beta = 0.653$, $p < 0.01$) are both positively and significantly associated with adoption. The pattern persists across all analyses; as Table 2 illustrates, ICT service exports never approach statistical significance once controls are introduced, suggesting that the apparent relationship observed in univariate models is largely driven by underlying differences in economic development and digital infrastructure rather than by ICT sector strength itself. The

substantive implications here are most visible when considering the predicted probabilities of the core control model: moving ICT service exports from 0 to 50 per cent of total service exports shifts the predicted probability of adoption from only 0.73 to 0.72 while holding other variables at their means—an effect that is both statistically and substantively negligible. A likelihood ratio test comparing the univariate and core controls regressions confirms that the addition of GDP, population, and broadband significantly improves model fit ($\chi^2 = 39.37$, $p < 0.001$), and the AIC declines from 153.8 to 120.4, favouring the multivariate specification. Taken together, these results provide little evidence that ICT sector development independently predicts AI strategy adoption.

Table 2: Logistic Regression — AI Strategy Adoption

	Dependent variable:			
	AI Strategy Adopted			
	Univariate	Core Controls	ICT Measures	Human Capital
	(1)	(2)	(3)	(4)
ICT service exports (%)	0.042*	-0.002	-0.007	0.005
	(0.023)	(0.023)	(0.025)	(0.031)
High-tech exports (%)			0.001	
			(0.028)	
ICT goods exports (%)			0.014	
			(0.047)	
Log GDP per capita		0.725*	0.879	0.756
		(0.390)	(0.545)	(0.643)
Log population		0.653***	0.844**	0.266
		(0.162)	(0.371)	(0.232)
Broadband per 100		0.046	0.060*	-0.015
		(0.030)	(0.037)	(0.048)
Log patent applications			-0.172	
			(0.252)	
Tertiary enrolment (%)				0.023
				(0.018)

R&D expenditure (% GDP)				0.871
				(0.663)
Constant	0.260	-17.828***	-21.959**	-12.646*
	(0.274)	(4.668)	(9.149)	(7.443)
AIC	153.8	120.4	94.3	75.6
Observations	120	120	92	82
Log Likelihood	-74.896	-55.209	-39.130	-30.789

Note: Standard errors in parentheses.
* p < 0.1; ** p < 0.05; *** p < 0.01

OLS Regression

Among the 79 countries that have adopted a strategy and for which adoption year data is available, ICT service exports show a stronger univariate relationship with the timing of adoption than with adoption itself. A one-percentage-point increase in ICT service exports is associated with adoption approximately 0.039 years earlier ($\beta = -0.039$, $p < 0.05$), though the R^2 of 0.08 indicates that ICT exports alone explain very little of the cross-national variation in when countries chose to adopt a strategy.

Table 3: OLS Regression — Year of Strategy Adoption

	Dependent variable:			
	Year of Strategy Adoption			
	Univariate	Core Controls	+ ICT Measures	+ Human Capital
	(1)	(2)	(3)	(4)
ICT service exports (%)	-0.039**	-0.019*	-0.013	-0.013
	(0.015)	(0.011)	(0.011)	(0.011)
High-tech exports (%)			-0.021	
			(0.015)	
ICT goods exports (%)			0.028	
			(0.020)	

Log patent applications			-0.242**	
			(0.112)	
Log GDP per capita		-0.907***	-0.460	-0.518*
		(0.249)	(0.298)	(0.286)
Log population		-0.236***	0.074	-0.164*
		(0.080)	(0.179)	(0.084)
Broadband per 100		-0.029*	-0.017	-0.027
		(0.015)	(0.016)	(0.017)
Tertiary enrolment (%)				0.004
				(0.006)
R&D expenditure (% GDP)				-0.181
				(0.140)
Constant	2,020.950***	2,034.716***	2,026.215***	2,029.154***
	(0.252)	(2.915)	(4.753)	(3.515)
Observations	79	79	69	64
R^2	0.080	0.568	0.565	0.371
Adjusted R^2	0.068	0.545	0.515	0.305
Residual Std. Error	1.591 (df = 77)	1.112 (df = 74)	1.032 (df = 61)	1.040 (df = 57)
F Statistic	6.663** (df = 1; 77)	24.348*** (df = 4; 74)	11.311*** (df = 7; 61)	5.599*** (df = 6; 57)
Note:				Standard errors in parentheses.
				* p < 0.1; ** p < 0.05; *** p < 0.01

The introduction of core controls substantially alters this relationship. Once GDP per capita, population, and broadband penetration are included, the explanatory power of the model increases sharply ($R^2 = 0.57$). As the OLS results in Table 3 indicate, log GDP per capita ($\beta = -0.907$, $p < 0.01$) and log population ($\beta = -0.236$, $p < 0.01$) emerge as the strongest predictors of earlier adoption, while ICT service exports remain marginally significant but reduced in magnitude ($\beta = -0.019$, $p < 0.1$) and broadband penetration is also associated with earlier adoption ($\beta = -0.029$, $p < 0.1$). ICT service exports lose significance entirely once additional ICT

indicators or human capital variables are introduced. This pattern suggests that the apparent association between ICT sector development and earlier adoption is largely explained by broader structural factors, particularly economic development and digital connectivity.

Residuals-versus-fitted and quantile-quantile plots for the core control variables (Figures 1 and 2) reveal some departure from the normality assumption, with heavy tails in the Q-Q plot and discrete clustering in the residual structure — consistent with the bounded and lumpy nature of adoption years as an outcome variable, where many countries adopt in the same year.

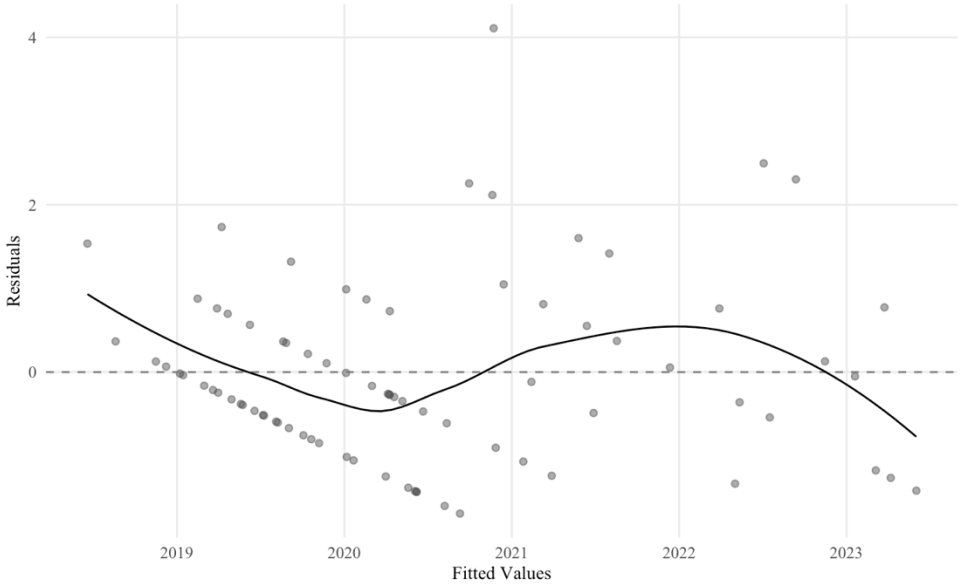

Figure 1. Residuals versus fitted values for the OLS regression of strategy adoption year on ICT service exports, log GDP per capita, log population, and broadband penetration (core controls specification, n = 79)

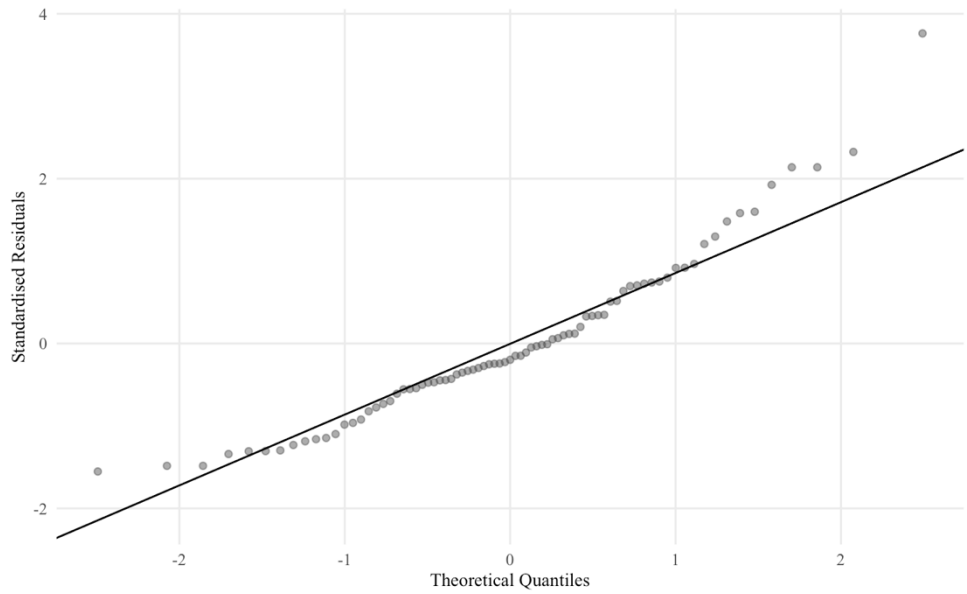

Figure 2. Normal Q-Q plot of standardised residuals for the OLS regression of strategy adoption year on the core controls specification (ICT service exports, log GDP per capita, log population, and broadband penetration; n = 79).

Poisson Regression

The count of AI-related bills passed into law presents a markedly different pattern. ICT service exports are not significantly associated with legislative output in the univariate model ($\beta = -0.010$, $p > 0.1$), but once core controls are introduced, the coefficient becomes negative and significant under default Poisson standard errors ($\beta = -0.031$, $p < 0.01$). However, the dispersion parameter for this model is 3.32, well above the 1.5 threshold indicative of overdispersion, meaning that default Poisson standard errors are likely too small. Re-estimating with HC1 robust standard errors substantially attenuates the precision of these estimates: the ICT services coefficient remains negative but is significant only at the 10% level (robust *SE* = 0.017), while GDP per capita ($\beta = 0.991$, robust *SE* = 0.420) and log population ($\beta = 0.477$, robust *SE* = 0.113) remain significant at conventional levels. Broadband penetration, significant under default *SE*s, is not distinguishable from zero once overdispersion is accounted for ($\beta = 0.030$, robust *SE* =

0.026). In multiplicative terms, each additional percentage point of ICT service exports remains associated with a 3.1 percent decrease in the expected count of bills, but the strength of this conditional association should be interpreted with caution. This pattern persists across all Poisson specifications, the full results of which appear in Table 4.

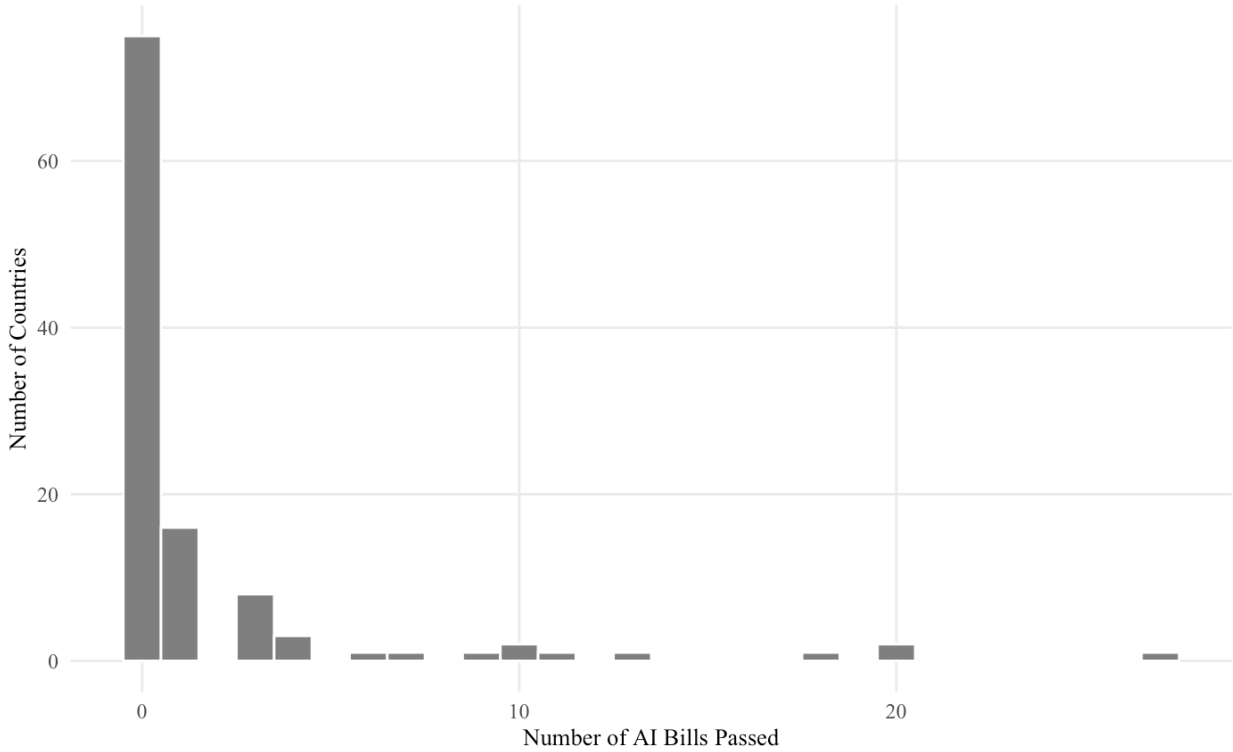

Figure 3. Distribution of AI-related bills passed into law across 113 countries. The modal count is zero, and the distribution is heavily right-skewed, motivating the use of Poisson regression.

Table 4: Poisson Regression — AI Bills Passed

	Dependent variable:				
	AI Bills Passed				
	Univariate	Core Controls	Core Controls (HC1)	+ ICT Measures	+ Human Capital
	(1)	(2)	(3)	(4)	(5)
ICT service exports (%)	-0.010	-0.031***	-0.031*	-0.033***	-0.045***
	(0.007)	(0.010)	(0.017)	(0.011)	(0.012)
High-tech exports (%)				0.0002	

				(0.009)	
ICT goods exports (%)				-0.001	
				(0.011)	
Log patent applications				0.053	
				(0.068)	
Log GDP per capita		0.991***	0.991**	0.772***	0.606***
		(0.174)	(0.420)	(0.220)	(0.205)
Log population		0.477***	0.477***	0.380***	0.388***
		(0.043)	(0.113)	(0.114)	(0.050)
Broadband per 100		0.030***	0.030	0.027***	0.021**
		(0.009)	(0.026)	(0.009)	(0.010)
Tertiary enrolment (%)					-0.003
					(0.004)
R&D expenditure (% GDP)					0.230***
					(0.082)
Constant	0.748***	-18.425***	-18.425***	-14.645***	-12.435***
	(0.098)	(2.104)	(5.160)	(3.320)	(2.565)
Dispersion (deviance/df)		3.32	3.32		
Observations	105	105	105	81	76
Log Likelihood	-377.673	-219.862	-219.862	-211.167	-205.792
Akaike Inf. Crit.	759.345	449.724	449.724	438.334	425.583

Note: Standard errors in parentheses. Column 3 uses HC1 robust SEs.
* p < 0.1; ** p < 0.05; *** p < 0.01

Robustness Checks and Extensions

To address the concern that certain variables might reflect the influence of strategy adoption—given that several countries adopted strategies as early as 2017— all specifications are re-estimated using a parallel baseline dataset with World Bank indicators drawn from 2016. The pattern of results is effectively unchanged: ICT service exports are weakly significant in univariate logistic specifications, but lose significance with core controls, GDP and population

remain the dominant predictors of adoption timing, and the Poisson models continue to show a negative conditional association between ICT service exports and legislative output, indicating that the core findings are robust to alternative temporal specifications and are unlikely to be driven by reverse causality.

Additional analyses substituting the IMF AI Preparedness Index and the ITU Internet Development Index for broadband penetration also produce consistent findings, with AI Preparedness reaching significance in the logistic model ($\beta = 13.903$, $p < 0.01$) but neither composite index altering the core result that ICT service exports do not independently predict strategy adoption.

Discussion

This paper asked whether countries with larger and more developed ICT sectors are more likely to adopt national AI strategies, do so at an earlier rate, and produce more AI-related policies and laws than countries with less developed ICT sectors. It hypothesised that ICT sector development—measured through the export orientation of a country's technology industries—would positively predict the likelihood of adoption, the timing of adoption, and the volume of AI-related legislation. The results offer a clear answer: the digital divide in AI governance is real, but it is not primarily an ICT-sector story.

The logistic regression models showed that ICT service exports are positively associated with strategy adoption when considered in isolation, but that this association vanishes entirely once GDP per capita, population, and broadband penetration are introduced as controls. The same pattern broadly holds across the OLS analyses, with one notable difference in the Poisson models, where the coefficient becomes negative and weakly significant under HC1 robust

standard errors, despite being non-significant in the univariate specification. What appears at first glance to be a technology-sector effect turns out to instead largely serve as a proxy for the broader structural conditions—economic wealth, country size, digital infrastructure—that independently and collectively drive both ICT sector development and governmental AI activity. The hypothesis that ICT exports would independently predict AI governance outcomes is, in short, not supported by the data. Countries with large ICT sectors do tend to adopt AI strategies, but not because of their ICT sectors per se—rather, because they tend also to be wealthy, populous, and well-connected, and it is these underlying conditions that do the explanatory work in these analyses.

This finding is broadly consistent with the cross-national technology diffusion literature. Comin and Hobijn (2004) documented a pattern of "trickle-down" diffusion across twenty-five technologies and twenty-three countries in which GDP per capita was the dominant predictor of adoption, and the present results suggest that AI governance follows a similar logic—not as a technology to be adopted, but as a policy response whose emergence depends on the same developmental prerequisites. It also aligns with Ali, Mustafa and Aysan's (2025) finding that IT infrastructure and R&D investment drive generative AI adoption cross-nationally, while poorly aligned government policies can hinder it. The Poisson result—that countries with larger ICT sectors may have passed fewer AI bills—may reflect a parallel dynamic: economies heavily oriented toward ICT exports may have less political incentive to regulate AI aggressively, or may rely on industry self-governance rather than legislative intervention.

The results also speak to the comparative AI strategy literature. Fatima, Desouza and Dawson (2020) documented significant variation in the content of national AI plans but did not attempt to explain why some countries adopted strategies and others did not. The present

analysis suggests that the answer lies less in the sophistication of a country's technology sector than in the general capacity of its state and economy—a conclusion that resonates with Fatima et al.'s (2022) subsequent finding that the resources required to implement AI strategies are principally technological infrastructure and human capital, both of which correlate more strongly with broad development than with sector-specific ICT orientation.

Several limitations must be considered. First, the cross-sectional design of this analysis cannot establish causality. Although the 2016 baseline robustness check helps to address reverse causality, the analysis cannot rule out omitted variables that might confound the relationship between development indicators and strategy adoption. Second, the OLS analysis conditions on having adopted a strategy, which introduces selection bias: the factors that predict *when* a country adopts likely differ from those that predict whether it does adopt a strategy, which likely biases the results. Third, the Poisson models exhibit substantial overdispersion ($\varphi = 3.32$). HC1 robust standard errors confirm that the negative conditional association between ICT exports and legislative output persists in direction but weakens substantially in significance (from $p < 0.01$ to $p < 0.1$), and broadband loses significance entirely. The negative ICT–bills association should therefore be treated as suggestive rather than definitive. Finally, "strategy adoption" is a coarse measure—it captures whether a document has been published, not whether it has been implemented, funded, or enforced in any meaningful manner. As Fatima et al. (2022) note, many national AI plans remain largely aspirational, and the present analysis cannot distinguish between strategies that represent genuine policy commitments and those that are primarily symbolic.

Despite these limitations however, the findings carry a practical implication. If the global AI governance gap is driven not by sector-specific ICT capacity, but by the foundational

conditions of economic development and digital infrastructure, then efforts to accelerate AI strategy adoption in the Global South are unlikely to succeed through narrowly targeted interventions—subsidising domestic software industries, for instance, or promoting ICT exports. The more productive path, these results suggest, runs through the kinds of investments that raise broadband penetration, strengthen institutional capacity, and build the general economic foundations on which both technology sectors and technology governance depend. In this sense, the global distribution of AI governance reflects not a frontier of technological specialisation, but a continuation of deeper structural inequalities in economic development and state capacity.

Works Cited

Ali, H., Mustafa, A. ul, & Aysan, A. F. (2025). Global adoption of generative AI: What matters most? *Journal of Economy and Technology*, *3*, 166–176. https://doi.org/10.1016/j.ject.2024.10.002

Caiazza, R. (2016). A cross-national analysis of policies affecting innovation diffusion. *The Journal of Technology Transfer*, *41*(6), 1406–1419. https://doi.org/10.1007/s10961-015-9439-2

Cazzaniga, M., Jaumotte, F., Li, L., Melina, G., Panton, A. J., Pizzinelli, C., Rockall, E., & Tavares, M. M. (2024). *Gen-AI: Artificial Intelligence and the Future of Work* (Nos. 2617–6750; IMF Staff Discussion Notes, Vol. 2024, Issue 001). International Monetary Fund. https://doi.org/10.5089/9798400262548.006

Comin, D., & Hobijn, B. (2004). Cross-country technology adoption: Making the theories face the facts. *Journal of Monetary Economics*, *51*(1), 39–83. https://doi.org/10.1016/j.jmoneco.2003.07.003

Demaidi, M. N. (2025). Artificial intelligence national strategy in a developing country. *AI & SOCIETY*, *40*(2), 423–435. https://doi.org/10.1007/s00146-023-01779-x

Fatima, S., Desouza, K. C., & Dawson, G. S. (2020). National strategic artificial intelligence plans: A multi-dimensional analysis. *Economic Analysis and Policy*, *67*, 178–194. https://doi.org/10.1016/j.eap.2020.07.008

Fatima, S., Desouza, K. C., Dawson, G. S., & Denford, J. S. (2022). Interpreting national artificial intelligence plans: A screening approach for aspirations and reality. *Economic Analysis and Policy*, *75*, 378–388. https://doi.org/10.1016/j.eap.2022.04.012

International Monetary Fund. (2024). *AI Preparedness Index (AIPI)* [Dataset]. International Monetary Fund. https://www.imf.org/external/datamapper/datasets/AIPI

International Telecommunication Union. (2024a). *Measuring digital development: The ICT Development Index 2024*. International Telecommunication Union. https://www.itu.int/dms_pub/itu-d/opb/ind/D-IND-ICT_MDD-2024-3-PDF-E.pdf

International Telecommunication Union. (2024b). *Measuring digital development: The ICT Development Index 2024* [Dataset]. International Telecommunication Union. https://datahub.itu.int/dashboards/idi/

Kergroach, S., & Héritier, J. (2025). Emerging divides in the transition to artificial intelligence. *OECD Regional Development Papers*. https://doi.org/10.1787/7376c776-en

Maslej, N., Fattorini, L., Perrault, R., Gil, Y., Parli, V., Kariuki, N., Capstick, E., Reuel, A., Brynjolfsson, E., Etchemendy, J., Ligett, K., Lyons, T., Manyika, J., Niebles, J. C., Shoham, Y., Wald, R., Walsh, T., Hamrah, A., Santarlasci, L., … Oak, S. (2025a). *The AI Index 2025 Annual Report* (AI Index Steering Committee, Institute for Human-Centered AI, Stanford University) [Annual Report]. Stanford University.

Maslej, N., Fattorini, L., Perrault, R., Gil, Y., Parli, V., Kariuki, N., Capstick, E., Reuel, A., Brynjolfsson, E., Etchemendy, J., Ligett, K., Lyons, T., Manyika, J., Niebles, J. C., Shoham, Y., Wald, R., Walsh, T., Hamrah, A., Santarlasci, L., … Oak, S. (2025b). *The AI Index 2025 Annual Report—Public Data* [Dataset]. AI Index Steering Committee, Institute for Human-Centered AI, Stanford University. https://drive.google.com/drive/folders/1AxxxL9-AsaeMdDKtTNHCR1KqEJTsHCod

Nhamo, G., Nhemachena, C., & Nhamo, S. (2020). Using ICT indicators to measure readiness of countries to implement Industry 4.0 and the SDGs. *Environmental Economics and Policy Studies*, *22*(2), 315–337. https://doi.org/10.1007/s10018-019-00259-1

Sajadieh, S., Fattorini, L., Perrault, R., Gil, Y., Parli, V., Santarlasci, L., Pava, J., Maslej, N., Altman, R., Brynjolfsson, E., Brodley, C., Clark, J., Dignum, V., Kumar, V., Landay, J., Lyons, T., Manyika, J., Niebles, J. C., Shoham, Y., … Weld, D. (2026a). *The AI Index 2026 Annual Report* (AI Index Steering Committee, Institute for Human-Centered AI, Stanford University) [Annual Report]. Stanford University.

Sajadieh, S., Fattorini, L., Perrault, R., Gil, Y., Parli, V., Santarlasci, L., Pava, J., Maslej, N., Altman, R., Brynjolfsson, E., Brodley, C., Clark, J., Dignum, V., Kumar, V., Landay, J., Lyons, T., Manyika, J., Niebles, J. C., Shoham, Y., … Weld, D. (2026b). *The AI Index 2026 Annual Report—Public Data* [Dataset]. AI Index Steering Committee, Institute for Human-Centered AI, Stanford University. https://drive.google.com/drive/folders/1zJTOg0iR0j5SijCwFutwWvDt143lW277?usp=sharing

Smuha, N. A. (2019). *From a "Race to AI" to a 'Race to AI Regulation'—Regulatory Competition for Artificial Intelligence* (SSRN Scholarly Paper No. 3501410). Social Science Research Network. https://doi.org/10.2139/ssrn.3501410

Ulnicane, I., Knight, W., Leach, T., Stahl, B. C., & Wanjiku, W.-G. (2021). Framing governance for a contested emerging technology:insights from AI policy. *Policy and Society*, *40*(2), 158–177. https://doi.org/10.1080/14494035.2020.1855800

United Nations Conference on Trade and Development, Press, L., Dumans, M.-E., Hamdi, M., Minges, M., Wolcott, P., Biggs, P., Kelly, T., & Lanvin, B. (2006). *THE DIGITAL DIVIDE REPORT: ICT DIFFUSION INDEX, 2005*. https://unctad.org/system/files/official-document/iteipc20065_en.pdf

World Bank. (2025). *World Development Indicators* [Dataset: Fixed broadband subscriptions per 100 people (IT.NET.BBND.P2); GDP per capita, PPP constant 2021 international $

(NY.GDP.PCAP.PP.KD); Government expenditure on education, % of government expenditure (SE.XPD.TOTL.GB.ZS); High-technology exports, % of manufactured exports (TX.VAL.TECH.MF.ZS); ICT goods exports, % of total goods exports (TX.VAL.ICTG.ZS.UN); ICT service exports, % of service exports BoP (BX.GSR.CCIS.ZS); Individuals using the Internet, % of population (IT.NET.USER.ZS); Mobile cellular subscriptions per 100 people (IT.CEL.SETS.P2); Patent applications, residents (IP.PAT.RESD); Population, total (SP.POP.TOTL); Research and development expenditure, % of GDP (GB.XPD.RSDV.GD.ZS); School enrolment, tertiary, % gross (SE.TER.ENRR)]. The World Bank. https://databank.worldbank.org/source/world-development-indicators